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Hardware

MKR Vidor 4000



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This tutorial refers to a product that has reached its end-of-life status.

Host a Web Server on the MKR Vidor 4000

Learn how to access your board through a browser on the same network.

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Introduction

In this tutorial, we will use the MKR Vidor 4000 board to set up a simple web server, using the **WiFinina** library. The web server will be used as an interface for our board, where we will create two buttons to remotely turn ON or OFF an LED.

This tutorial is a great starting point for any maker interested in making applications connected to the Internet.

Hardware & Software Needed

Arduino IDE ([online](#) or

[Help](#)

VidorPeripherals library.

MKR Vidor 4000 ([link to store](#)).

Generic LED

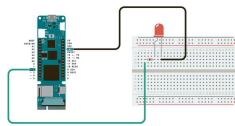
82 ohm resistor

Breadboard

Jumper wires

Circuit

Follow the wiring diagram below to connect the LED to the MKR Vidor 4000 board.



The circuit.

Let's Start

This tutorial barely uses any external hardware, except an LED that we will control remotely. However, the most interesting aspect lies in the library we are going to use: **WiFinina**. This library can be used for many different connectivity projects, where it allows us to connect to WiFi, make GET requests and - as we will explore in this tutorial - to create a web server.

1. First, let's make sure we have the drivers installed. If we are using the Cloud Editor, we do not need to install anything. If we are using an offline editor, we need to install it manually. This can be

Board > Board Manager... Here we need to look for the **Arduino SAMD boards (32-bits Arm® Cortex®-M0+)** and install it.

2. Now, we need to install the library needed. If we are using the Cloud Editor, there is no need to install anything. If we are using an offline editor, simply go to **Tools > Manage libraries..**, and search for **WiFinina** and install it.

3. We will also need to add an additional library, called **VidorPeripherals**. Search for it in the manage libraries and install it. The author of this library is Martino Facchin.



Installing the library.

4. We can now take a look at some of the core functions of this sketch:

`char ssid[] = ""` - stores network name.

`char pass[] = ""` - stores network password.

`WiFi.begin(ssid, pass)` connects to Wi-Fi with credentials.

`WiFiServer server(80)` - creates a server that listens for incoming connections on the specified port.

`WiFiClient client` - creates a client that can connect to a specified internet IP address.

`server.begin()` - tells the

`client.connected` - checks for connected clients.

`client.available` - checks for available data.

`client.read` - reads the available data.

`client.print()` - print something to the client (e.g. html code).

`client.stop()` - closes the connection.

The sketch can be found in the snippet below. It is very similar to the sketch found in **File > Examples > WiFinina > SimpleWebServer**, with only minor modifications.

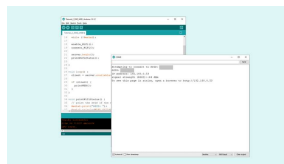
Upload the code to the board, and make sure the right board and port are selected.



```
1 #include <WiFiNINA.h>
2
3 char ssid[] = "";
4 char pass[] = "";
5 int keyIndex = 0;
6 int status = WL_IDLE
7 WiFiServer server(80
8
9 WiFiClient client =
10
11 int ledPin = 2;
12
13 void setup() {
14     Serial.begin(9600)
15     pinMode(ledPin, OU
16     while (!Serial);
17
18     enable_WiFi();
19     connect_WiFi();
20
21     server.begin();
22     printWifiStatus();
23
24 }
25
26 void loop() {
27     client = server.av
28
```

Testing It Out

Once we have successfully uploaded the code to the board, open the Serial Monitor and it should look like the following image:



Serial monitor output.

Copy the IP address and enter it in a browser. Now, we should see a very empty page with two links at

to turn the LED on" and "Click here to turn the LED off".



The web interface.

When interacting with the links, you should see the LED, connected to pin 2, turn on and off depending on what you click. Now we have successfully created a way of interacting with our MKR Vidor 4000 remotely.

Troubleshoot

If the code is not working, there are some common issues we might need to troubleshoot:

- We have not updated the latest firmware for the board.

- We have not installed the Board Package required for the board.

- We have not installed the **WiFiNINA** library.

- We have entered the SSID and PASS incorrectly: remember, it is case sensitive.

- We have not selected the right port to upload: depending on what computer we use, sometimes the board is duplicated. By simply restarting the editor, this issue can be solved.

Conclusion

In this tutorial, we learned how to create a basic web interface from scratch. We learned how to control an LED remotely, and how to display the value of an analog pin in the browser as well. Using this example, we can build much more complex projects, and if you are familiar with both HTML and CSS, you can create some really cool looking interfaces! If you are new to HTML and CSS, there are plenty of online guides that can guide you, you can visit [w3schools](#) or [codecademy](#) for many tips and tricks.

Tip: Check out [fontawesome](#) to get access to thousands of free icons that you can customize your local web server with!

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